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Chapter 3

Convex Optimization in Machine Learning

Mathematical Modeling

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$$\min \|Ax - b\|_2^2 = \sum_{i=1}^k (a_i^T x - b_i)^2,$$

where $A \in \mathbb{R}^{k \times n}$ ($k \geq n$), $b \in \mathbb{R}^k$, a_i^T are rows of A , and $x \in \mathbb{R}^n$.

Solving least-squares problems

- Analytical solution: $x^* = (A^T A)^{-1} A^T b$.
- Reliable and efficient algorithms and software.
- Computation time proportional to $n^2 k$.
- A mature technology.

- Basis for regression analysis, optimal control, and many parameter estimation and data fitting methods.
- Easy to recognize.
- A few standard techniques increase flexibility
 - *Including weights*

$$\min \sum_{i=1}^k w_i (a_i^T x - b_i)^2.$$

- *Adding regularization terms*

$$\min \sum_{i=1}^k (a_i^T x - b_i)^2 + \rho \sum_{j=1}^n x_j^2.$$





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General statement

In a fitting problem, we are given data

$$(u_1, y_1), (u_2, y_2), \dots, (u_m, y_m)$$

with $u_i \in D$ and $y_i \in \mathbb{R}$, and seek a function $f \in \mathcal{F}$ that matches this data as closely as possible. For example in least-squares fitting we consider the problem

$$\min \sum_{j=1}^m (f(u_i) - y_i)^2,$$

which is a simple least-squares problem in the variable x .

We can add a variety of constraints

- inequalities that must be satisfied by f at various points,
- constraints on the derivatives of f ,
- monotonicity constraints,
- or moment constraints.



Polynomial fitting

Given data $u_1, u_1, \dots, u_m \in \mathbb{R}$ and $v_1, v_2, \dots, v_m \in \mathbb{R}$, we need to approximately fit a polynomial of the form

$$p(u) = x_1 + x_2 u + x_3 u^2 \dots + x_n u^{n-1}$$

to the data.

For each x we form the vector of errors,

$$e = (p(u_1) - v_1, p(u_2) - v_2, \dots, p(u_m) - v_m).$$

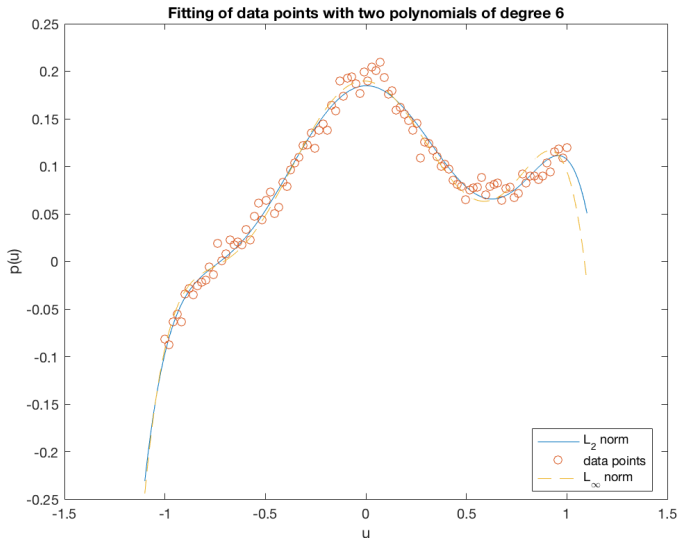
To find the polynomial that minimizes the norm of the error, we solve the norm approximation problem

$$\min \|e\| = \|Ax - v\|,$$

where $x \in \mathbb{R}^n$, $A_{ij} = u_i^{j-1}$, for $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$

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Polynomial fitting



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- The problem of estimating underlying trends in time series data arises in a variety of disciplines.
- The l_1 trend filtering method produces trend estimates x that are piecewise linear from the time series y .

Optimization form

Given a time series data y , estimate x by solving

$$\min \frac{1}{2} \|y - x\|^2 + \lambda \|Dx\|_1,$$

where D is the second difference matrix, with rows $[0 \dots -1 \ 2 \ -1 \dots 0]$, $\lambda > 0$ is the regularization parameter.



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Trend filter: time series analysis



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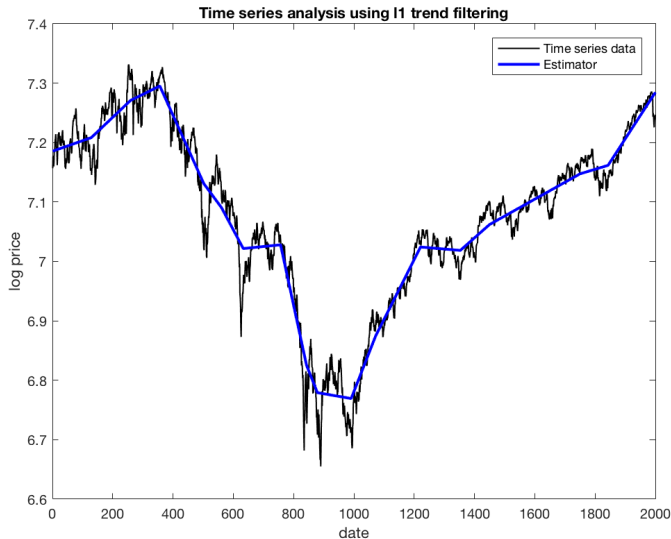
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A statement

A classification problem has two types of variables,

- $x_i \in \mathbb{R}^d$: the vector of observations (features) in the world,
- $y_i \in \mathbb{R}^d$: state (class) vector of the world.

The set of all samples in this world (or dataset) constructs the observation set $X \in \mathcal{R}^{n \times d}$ and state set $Y \in \mathbb{R}^{n \times d}$. The task of machine learning algorithm for classification is to find a mapping $X \rightarrow Y$ that achieves smallest number of misclassification.

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- *Propose a parametric family of decision boundaries, then pick the element in this family that produces the best classifier.*
- Take our decision boundary between two classes as a hyperplane, such that $w^T x + b = 0$,
 - w being the normal to the plane and b being the *bias term*.
- The decision function can be expressed as

$$f^*(x) = \begin{cases} 0, & \text{if } w^T x + b > 0, \\ 1, & \text{if } w^T x + b < 0. \end{cases}$$

The decision boundary

- Normal vector: w ; distance to origin: $d = b / \|w\|$.
- distance to a data point x_i : $|w^T x_i + b| / \|w\|$.



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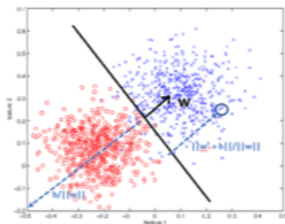
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Linear discriminant learning



The decision function

- Use labels $y \in \{-1, 1\}$ instead of $y \in \{0, 1\}$

$$f^*(x) = \begin{cases} -1, & \text{if } w^T x + b > 0, \\ 1, & \text{if } w^T x + b < 0, \end{cases}$$

then $f^*(x) = \text{sign}(w^T x + b)$.

A simple necessary and sufficient condition for a given training set $D = (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, is *linearly separable*, i.e.,

$$y_i(w^T x_i + b) > 0, \text{ for } i = 1, 2, \dots, n. \quad (1)$$



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Question arising when dealing with nonlinear problems

- Trying to fit a nonlinear model? – not easy!
 - Idea: keep using the linear model
 - Map the problem a new (higher-dimensional) space (*called the feature space*) by doing a nonlinear transformation using suitably chosen basis functions;
 - Apply a linear model in *the feature space*.
 - Learning a linear boundary in a higher dimensional space will be equivalent to learning a nonlinear boundary in the current space.
- Known as the **kernel trick**.

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- If we introduce a mapping $\phi : X \rightarrow Z$ such that $\dim(Z) > \dim(X)$, we can achieve linear separability for some $k = \dim(Z)$.
- Instead of applying this mapping to each data point as $\phi(x)$, we can leverage the dot-product form

$$y_i \left(\sum_{j=1}^n \alpha_j y_j x_j^T x_i + b \right) \quad (2)$$

$$y_i \left(\sum_{j=1}^n \alpha_j y_j \phi(x_j)^T \phi(x_i) + b \right). \quad (3)$$

Then we apply the kernel trick, $K(x, z) = \phi(x)^T \phi(z)$.

- Some formulas for $K(x, z)$
 - ① $K(x, z) = x^T z$, linear kernel.
 - ② $K(x, z) = e^{-\frac{\|x-z\|^2}{\delta}}$, Gaussian kernel.
 - ③ $K(x, z) = (1 + x^T z)^k$, polynomial kernel.



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Support Vector Machines (SVM)

- SVMs go one step further from the necessary and sufficient condition given by (1) and works on the *margin* defined by the distance from the boundary to the closest point,

$$\gamma = \min_i \frac{|w^T x_i + b|}{\|w\|}.$$

- Maximizing γ is ill-defined, since γ does not change if both w and b are scaled by a factor λ . We then need some sort of normalization,
 - this can be done by arbitrarily selecting some normalization on w , e.g. $\|w\| = 1$
- The SVM algorithm works with a more convenient normalization and makes $|w^T x + b| = 1$ for the closest point, i.e. $\min_i |w^T x_i + b|$ under $\gamma = \frac{1}{\|w\|}$.

SVM model

$$\begin{aligned} \min_{w,b} \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & y_i(w^T x_i + b) \geq 1, \forall i. \end{aligned}$$



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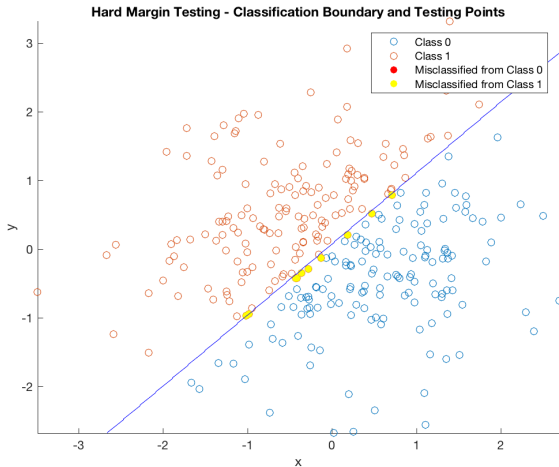
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SVM – hard margin

$$\begin{aligned} \min_{w,b} \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & y_i(w^T x_i + b) \geq 1, \forall i. \end{aligned}$$



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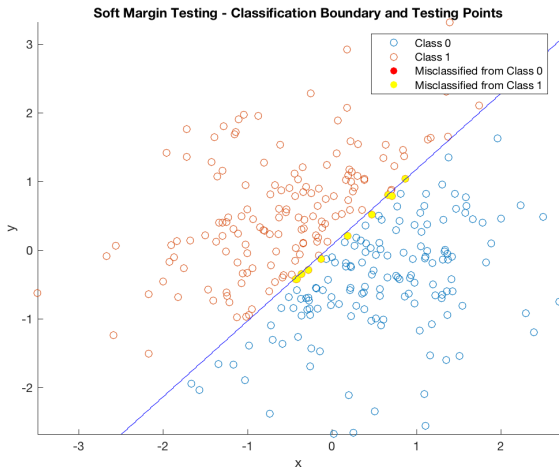
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SVM – soft margin

$$\begin{aligned} \min_{w,b,\xi} \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & y_i(w^T x_i + b) \geq 1 - \xi_i, \forall i, \\ & \xi_i \geq 0, \forall i. \end{aligned}$$



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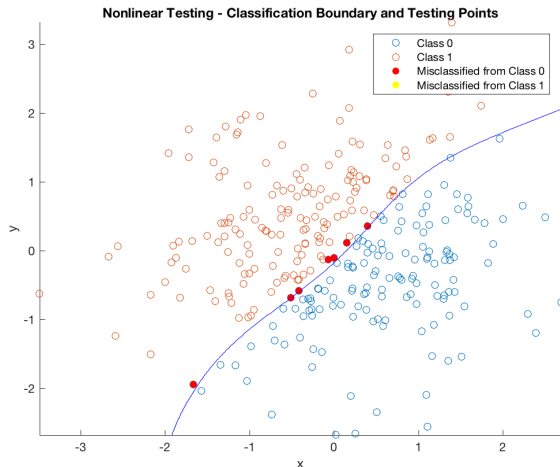
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SVM – nonlinear classification with kernels

Using Gaussian kernel: $K(x, z) = e^{-\frac{\|x - z\|^2}{\delta}}$.



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- original



- optical blur



- motion blur



- spatial quantization (discrete pixels)



- additive intensity noise



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- A general noisy (including noise, blur, inpainting) image $\mathbf{y}$, of an original image $\mathbf{x}$, can be modeled

$$\mathbf{y} = \mathbf{B}\mathbf{x} + \mathbf{n},$$

where

- $\mathbf{B}$: a direct operator represented under matrix form,
- $\mathbf{n}$: noise.
- As common, images are adopted by vector notation
 - The pixels on an $M \times N$ image are stored as an (NM) -vector.
 - If the number of elements of $\mathbf{x}$ is n then $\mathbf{x} \in \mathbb{R}^n$, while $\mathbf{y} \in \mathbb{R}^m$ (m and n can be different).

Remark

The problem of estimating $\mathbf{x}$ from $\mathbf{y}$ is *ill-posed*. This inverse problem can then be solved by adopting a some sort of regularization.



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Unconstrained formulation

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{B}\mathbf{x} - \mathbf{y}\|_2^2 + \tau \phi(\mathbf{x}),$$

where

- $\phi : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ is the regularizer.
- τ is the regularization parameter.

Lagrangian (constrained form)

$$\begin{aligned} \min_{\mathbf{x}} \quad & \phi(\mathbf{x}) \\ \text{s.t.} \quad & \|\mathbf{B}\mathbf{x} - \mathbf{y}\| \leq \epsilon. \end{aligned}$$

Regularization functions

- $\phi(\mathbf{x}) = \|\mathbf{x}\|_1 = \sum_{i=1} |x_i|$: basis pursuit denoising (BPD).
- $\phi(\mathbf{x}) = \varphi(\mathbf{D}\mathbf{x})$: **total-variational** regularization.



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Image restoration: inpainting and noise by unconstrained model



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Original



Missing Samples - 40%



Restored Image

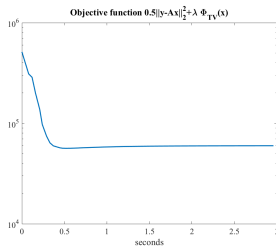
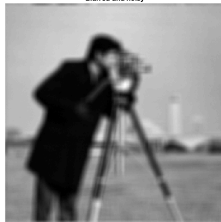


Image restoration: blurred by unconstrained model

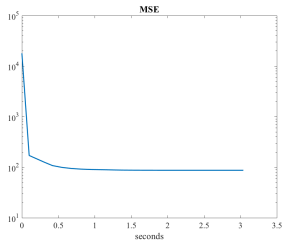
Original



Blurred and noisy



Estimated



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Image restoration: blurred and noise by constrained model



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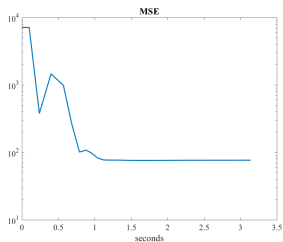
Original



Blurred and noisy



Constrained model result



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